2026-07-20 The infrastructure of the project

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Outline

Introduction to the Revelio data

- We currently have access to the Revelio lab data as of May 2025.
  - Job postings are from two sources: LinkedIn and Indeed.
  - Most job postings start from 2021, and end on May 15 2025.
  - Some Indeed job postings start earlier, but there are more LinkedIn postings in absolute numbers (at least for the 3 industries I will show you later).
- The datasets are stored on the Microsoft Fabric platform. Most data extraction, transformation, and computation tasks will be conducted within the platform.
- An extremely useful official guide to the data is here: link.
- The companion data guide when we purchased the data is in DropBox folder: "`data/a_raw_data/A_Revelio/Documents/`". It has more version-specific information compared to the official guide webpage in the last bullet point.

7 datasets of interest

- Firm side:
  - Company reference: One row per company, with company's baseline information about company ID, geography, industry, ...
  - Job postings from Indeed: One row per job posting, with company ID, posting/removal date, standardized job categories, geography, full job advertisements, estimated salary, ...
  - Job postings from LinkedIn: One row per job posting, with the same variables.[1]
- User side:
  - User positions: One row per reported employment spell, with company ID, start/end date, standardized job categories, geography, estimated salary, ...
  - User education: One row per reported education experience, with university/degree information, start/end date, ...
  - User demography and age: One row per user, with (estimated) age, race, gender, ...
- Random samples ($N = 1000$) are stored in DropBox folder "`data/a_raw_data/A_Revelio/`" as parquet files.
- Data description markdown files are in DropBox folder "`data/a_raw_data/A_Revelio/Descriptions/`". I list basic information about the dataset, e.g., unit of observation, variable names, and labels there.

[1] One job posting could show up in both sources, and they will be assigned with different job posting ID.

Outline

3 parts of the paper

Clustering $\Longrightarrow$ Validation and interpretation $\Longrightarrow$ Implications

1. Cluster firms based on their job postings and new hires' career trajectories.
2. Validate that our data-driven measure indeed has a reasonable economic interpretation.
   - For example, less experimentation in markets with high firing costs or labor market regulation.
3. Correlate with broader labor market implications.

The 3 parts are tightly linked.

- We need to pre-specify validation tests that will be used to decide which clustering algorithm works best.
  - Ideally, these tests should have solid economic theoretical foundation.
- We need to decide very very carefully which variables will be used to train models and classify firms, which variables will be used in validation, and which variables will be used for the implications.
  - For example, if we use new hires' objective productivity (like patents) in the clustering of firms (which of course, we shouldn't), then we cannot say anything about our measured firms' hiring practices on worker productivity, because it is already used to classify companies.
  - Another example: if we use countries' or states' firing costs to classify firms, then we cannot use this as a validation exercise to show that our measure makes sense.

Estimated timeline

Week	End date	Main task
1-2	2026-07-31	Get familiar with the data. Design a robust algorithm to match new hires to job postings.
3	2026-08-07	Design the textual algorithm to clean the job advertisements and extract variables.
4	2026-08-14	Extract different variables from employees' profiles.
5-6	2026-08-21	Test different clustering algorithms to classify firms by their hiring practices.
7-8	2026-09-04	Validate different clustering algorithms.

Tasks that need additional manual inputs

Steps that perhaps need some additional manual inputs:

- Validating our algorithm to match new hires to job postings produces reasonable matches.
  - Sample selection issue here: There is no way to match a new software engineer at Google to the corresponding job posting – so many job postings with identical job title: "software engineer".
  - We can only match new hires to postings in situation such as: a university hires 1-2 new Econ PhDs from the market.
  - This means that we will unnecessarily impose endogenous restrictions to the firm size.
- Validating our textual algorithm to extract variables from job advertisements is correct.
  - Some "hard" indicators (like the education/work experience, whether there is probation, the mention of training/mentoring, ...) have "correct" values, we need to check if we can extract right values from job advertisements.[2]
  - Some "soft" indicators (like how detailed the description of responsibilities/tasks, the mention of company culture, ...), we may need some topic modeling techniques from NLP, e.g., LDA, BERT, ...

[2] Admittedly, I don't have a very clear idea about how to extract these variables. My guess is that feeding the full job advertisements into AI chats with clearly specified prompts may be the easiest way to do this.

Outline

The definition of the 3 industries of interest

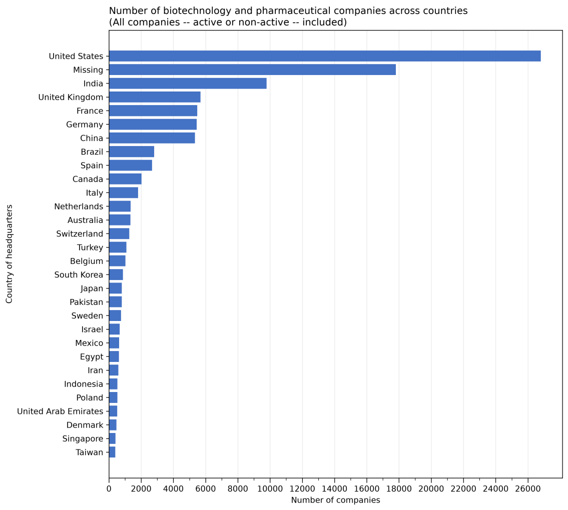
Revelio offers standardized industry codes, I will select companies of potential interest with the `rics_k400` variable and offer some baseline summary statistics in the final part of this document.

- BioPharm: `rics_k400` is either
  - Biotechnology and Life Sciences,
  - Pharmaceutical Manufacturing,
  - Pharmaceuticals.
- Automation: `rics_k400` is either
  - Automation and Robotics Equipment,
  - Automation and Robotics Solutions,
  - Industrial Automation and Tools,
  - Industrial Automation and Machinery.
- Electronics: `rics_k400` is either
  - Semiconductor and Electronics Manufacturing,
  - Electronic Components and Instrumentation,
  - Electronic Components and Equipment Manufacturing,
  - Electronics and Electrical Equipment Manufacturing,
  - Electronic Systems and Components.

Some comments on computation

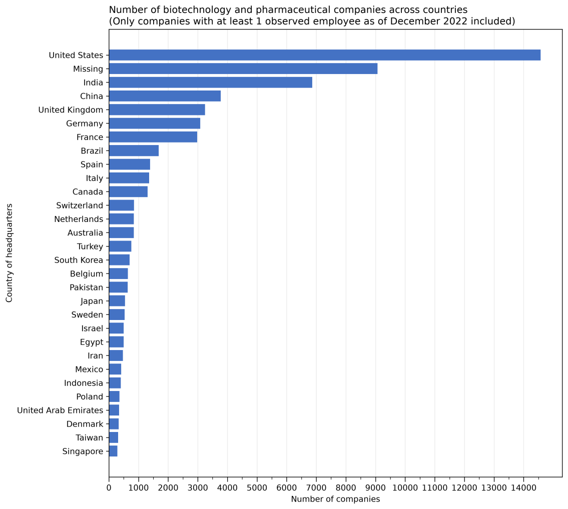
- From a computation perspective, the following analysis is more feasible/efficient – we first impose sample restrictions to the full data (e.g., only companies in certain industries), and then do analysis on this small subset of firms.
  - The size is moderately large: e.g., the dataset of LinkedIn job postings by all Electronics companies is about 18 GB.
  - It is feasible to analyze the data in one's own laptop after extracting only the relevant subset of data on the cloud platform.
- However, any analysis that needs to run on the full data will be very slow and computationally expensive.
  - For example, if we want to know the industry distribution among all firms in the dataset, this will be a very time-consuming task and can only run on the Fabric platform.

Number of companies across countries



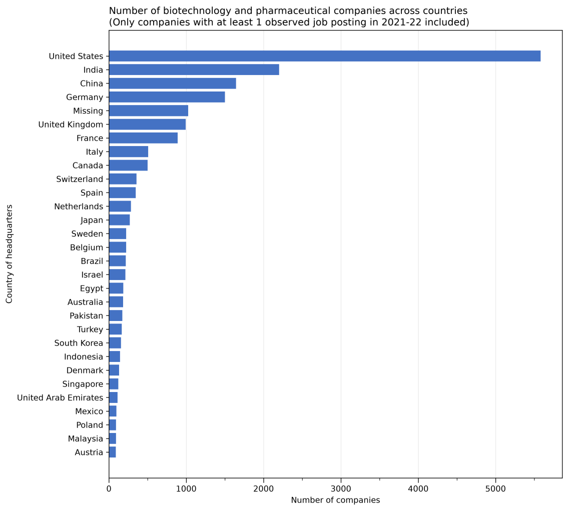
- In total, there are 110,760 companies in the BioPharm industry. Top 5: US, India, UK, France, Germany.
- However, not all of these firms are still active today. When firms exit, they don't necessarily update the company page in LinkedIn.

Number of active companies (defined as observed employee) across countries



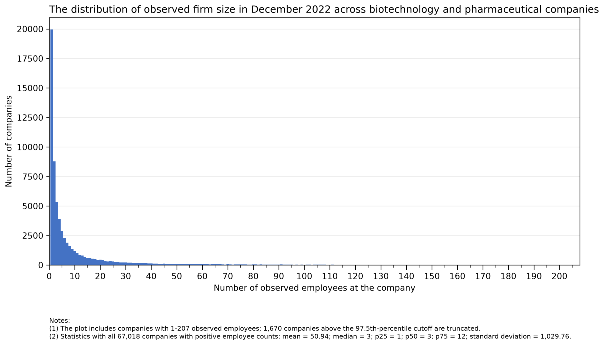
- There are 67,018 (60.5%) active companies (in the sense that there is at least 1 observed employee in LinkedIn profile in December 2022) in the BioPharm industry. Top 5: US, India, China, UK, Germany.

Number of active companies (defined as observed postings) across countries



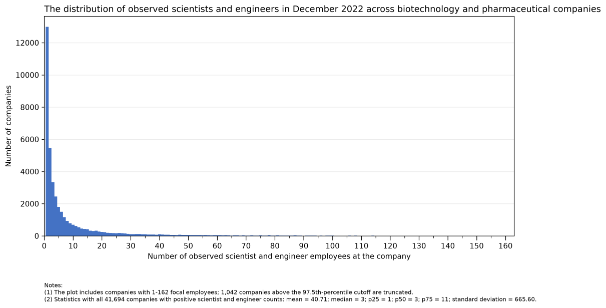
- There are 20,483 (18.5%) active companies (in the sense that there is at least 1 observed job posting in years 2021-2022) in the BioPharm industry. Top 5: US, India, China, Germany, UK.

Distribution of observed company size



- Among 67,018 companies with at least 1 observed employee in LinkedIn profile in December 2022, the mean employee count is 50.94, median is 3, p75 is 12, with a long right tail (std is 1029.76).
- Issues with this employee count:
  - Non-reporting: Company employees may not be active in LinkedIn.
  - Non-updating: When a LinkedIn user stops updating his profile, there will be no end date for his last employment spell, and he will be counted as the company's employee forever – no matter whether the company is still active or not.

Distribution of observed scientist and engineer employment across companies across companies

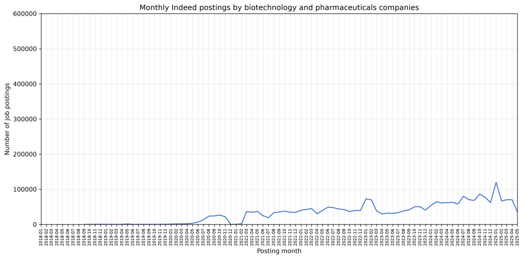


- Revelio offers standardized occupation variables. I will focus on only those employees whose `job_category`[3] are either Scientist or Engineer.
- Among 41,694 companies with at least 1 observed scientist or engineer in LinkedIn profile in December 2022, the mean focal employee count is 40.71, median is 3, p75 is 11, with a long right tail (std is 665.60).

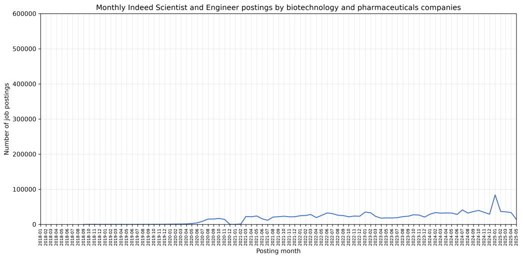
[3] This is the coarsest occupation variable, with only 7 broad categories: Engineer, Sales, Finance, Admin, Operations, Marketing, Scientist.

The number of job postings across time: Some weird patterns in the data?

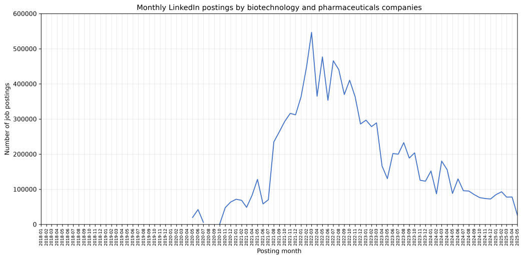
(a) Indeed, all job postings



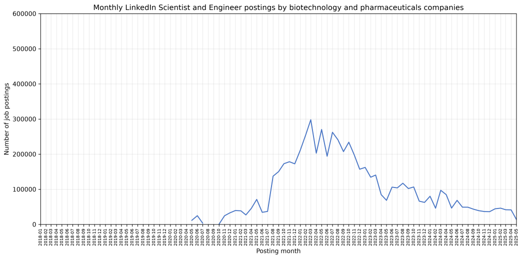
(b) Indeed, scientists and engineers



(c) LinkedIn, all job posting



(d) LinkedIn, scientists and engineers

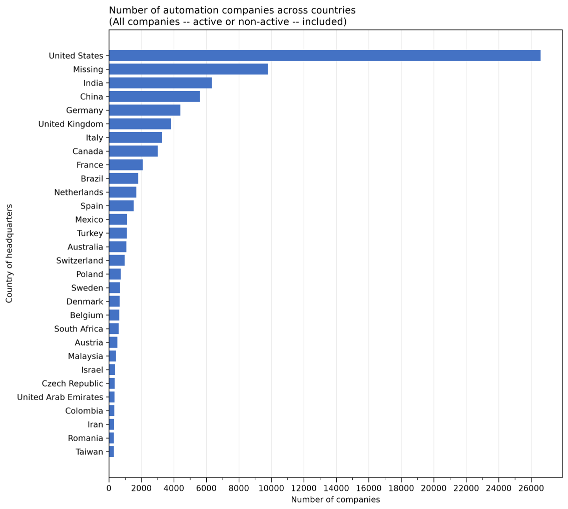


Some implications from the observed patterns

- The number of active companies (defined as those we can observe at least one job posting in years 2021-2022) within the BioPharm industries is around 20,000.
  - The country distribution is very uneven – most of them are in countries like US, UK, Germany, China, India.
  - After matching these job postings to the corresponding new hires, the sample size may be reduced further.
  - So we shouldn't expect an extremely large number of firms in the final sample used for clustering.
- The observed company size distribution is very right-skewed.
  - As I have said before, it is not realistic to match a new software engineer at Google to the corresponding job posting – so many job postings with identical job title: "software engineer".
  - Therefore, our matching algorithm (from new hires to job postings) introduces some endogenous sample selection based on firm size.
  - Our final company sample in the clustering exercise will be quite selective – they may not have the same right-skewed distribution as the observed company size distribution.
- The number of job postings reached a peak around January 2022 is very weird.
  - What is even more weird is the quick and sizeable decline in job postings after the peak (continuing until the end of our data).
  - This pattern also shows up in the other 2 industries below, not only the pharmaceutical industry.
  - I will check the codes and the data to see if there is any issue.

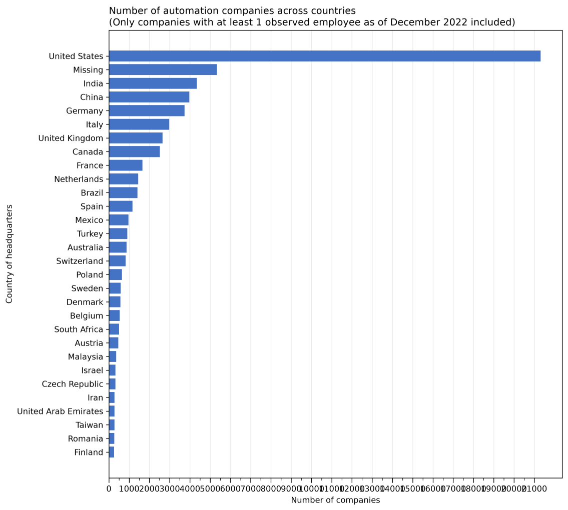
Outline

Number of companies across countries



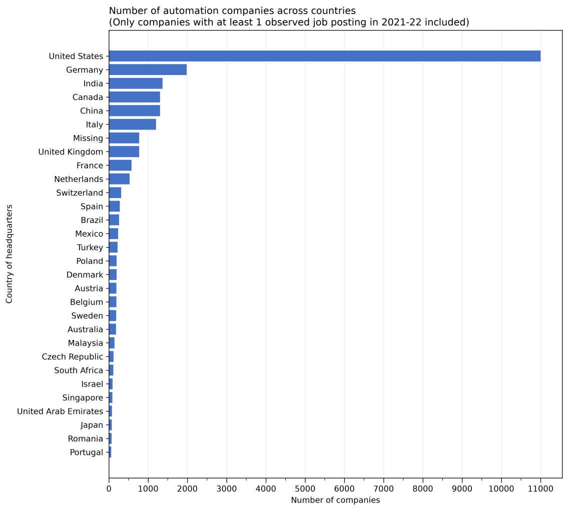
- In total, there are 88,390 companies in the Automation industry. Top 5: US, India, China, Germany, UK.
- However, not all of these firms are still active today. When firms exit, they don't necessarily update the company page in LinkedIn.

Number of active companies (defined as observed employee) across countries



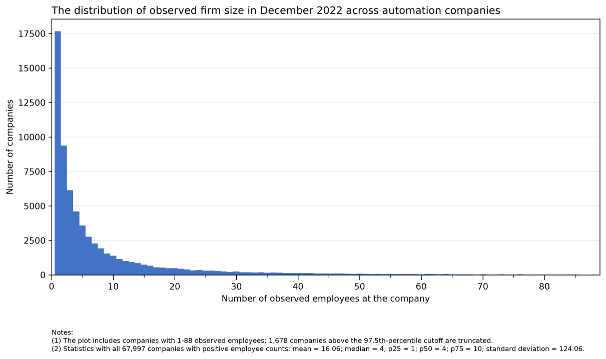
- There are 67,997 (76.9%) active companies (in the sense that there is at least 1 observed employee in LinkedIn profile in December 2022) in the Automation industry. Top 5: US, India, China, UK, Italy.

Number of active companies (defined as observed postings) across countries



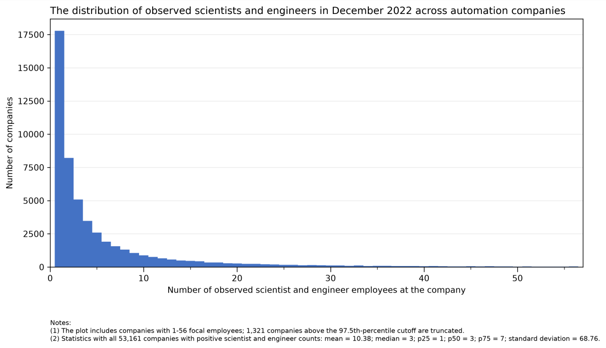
- There are 25,335 (28.7%) active companies (in the sense that there is at least 1 observed job posting in years 2021-2022) in the Automation industry. Top 5: US, Germany, India, Canada, China.

Distribution of observed company size



- Among 67,997 companies with at least 1 observed employee in LinkedIn profile in December 2022, the mean employee count is 16.06, median is 4, p75 is 10, with a long right tail (std is 124.06).
- Issues with this employee count:
  - Non-reporting: Company employees may not be active in LinkedIn.
  - Non-updating: When a LinkedIn user stops updating his profile, there will be no end date for his last employment spell, and he will be counted as the company's employee forever – no matter whether the company is still active or not.

Distribution of observed scientist and engineer employment across companies across companies

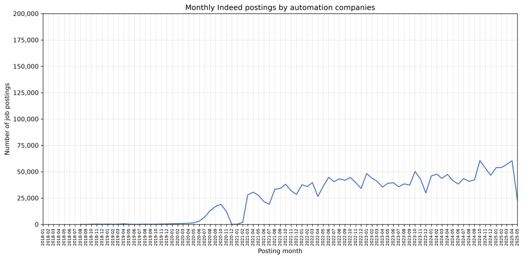


- Revelio offers standardized occupation variables. I will focus on only those employees whose `job_category`[4] are either Scientist or Engineer.
- Among 53,161 companies with at least 1 observed scientist or engineer in LinkedIn profile in December 2022, the mean focal employee count is 10.38, median is 3, p75 is 7, with a long right tail (std is 68.76).

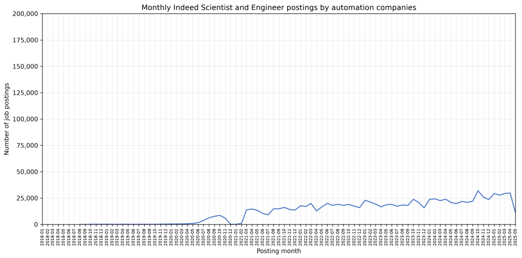
[4]This is the coarsest occupation variable, with only 7 broad categories: Engineer, Sales, Finance, Admin, Operations, Marketing, Scientist.

The number of job postings across time: Some weird patterns in the data?

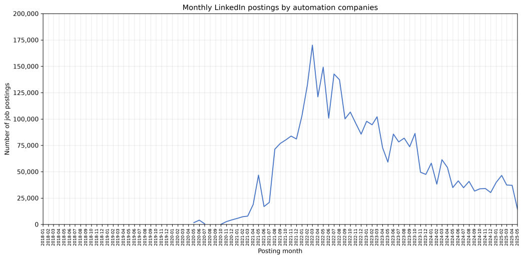
(a) Indeed, all job postings



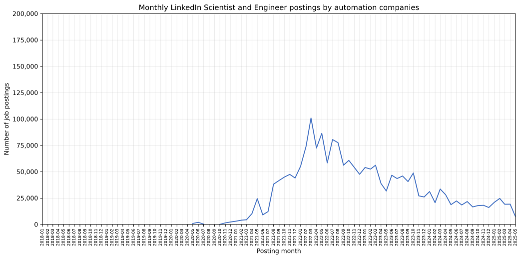
(b) Indeed, scientists and engineers



(c) LinkedIn, all job posting

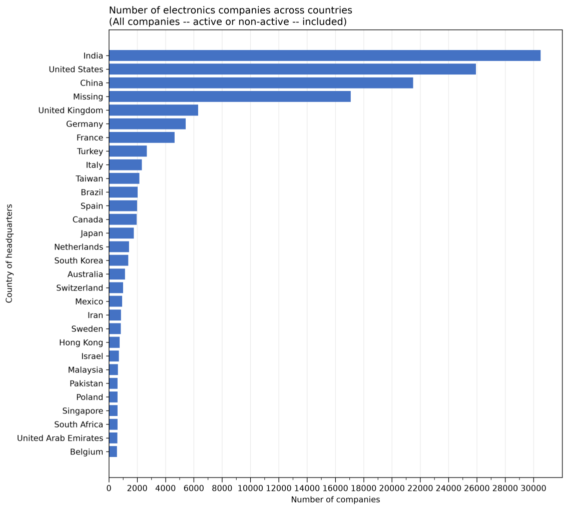


(d) LinkedIn, scientists and engineers



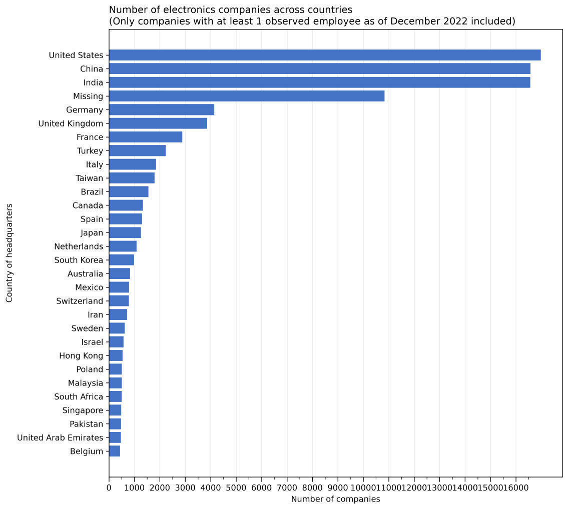
Outline

Number of companies across countries



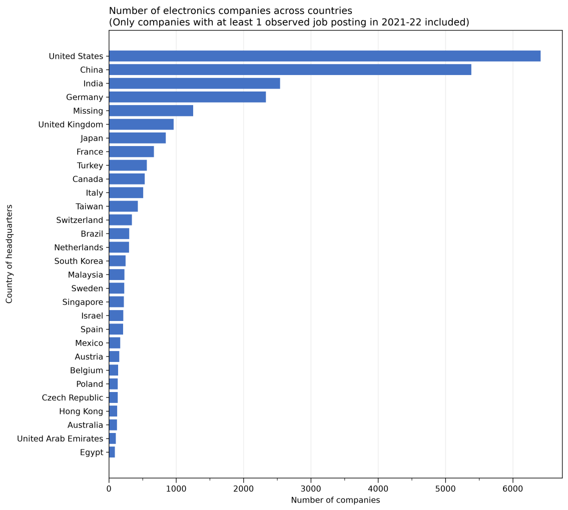
- In total, there are 151,266 companies in the Electronics industry. Top 5: India, US, China, UK, Germany.
- However, not all of these firms are still active today. When firms exit, they don't necessarily update the company page in LinkedIn.

Number of active companies (defined as observed employee) across countries



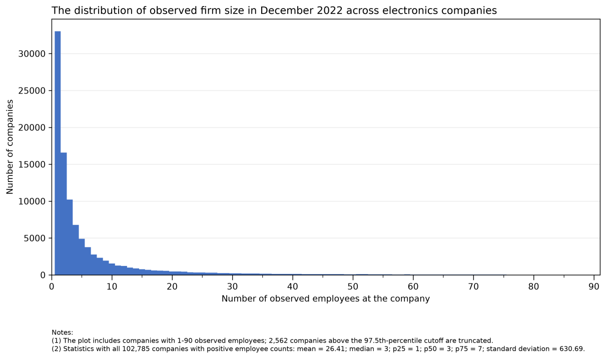
- There are 102,785 (67.9%) active companies (in the sense that there is at least 1 observed employee in LinkedIn profile in December 2022) in the Electronics industry. Top 5: US, China, India, Germany, UK.

Number of active companies (defined as observed postings) across countries



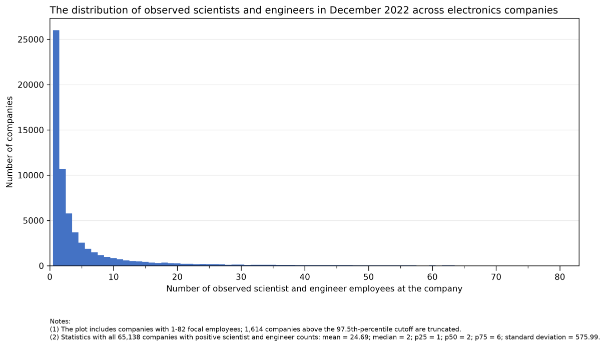
- There are 27,586 (18.2%) active companies (in the sense that there is at least 1 observed job posting in years 2021-2022) in the Electronics industry. Top 5: US, China, India, Germany, UK.

Distribution of observed company size



- Among 102,785 companies with at least 1 observed employee in LinkedIn profile in December 2022, the mean employee count is 26.41, median is 3, p75 is 7, with a long right tail (std is 630.69).
- Issues with this employee count:
  - Non-reporting: Company employees may not be active in LinkedIn.
  - Non-updating: When a LinkedIn user stops updating his profile, there will be no end date for his last employment spell, and he will be counted as the company's employee forever – no matter whether the company is still active or not.

Distribution of observed scientist and engineer employment across companies across companies

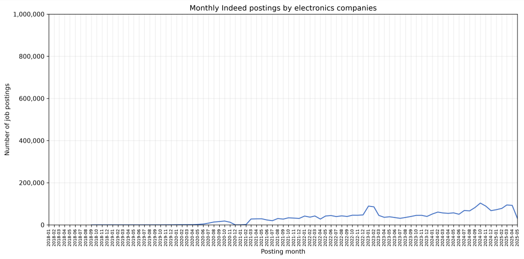


- Revelio offers standardized occupation variables. I will focus on only those employees whose `job_category`[5] are either Scientist or Engineer.
- Among 65,138 companies with at least 1 observed scientist or engineer in LinkedIn profile in December 2022, the mean focal employee count is 24.69, median is 2, p75 is 6, with a long right tail (std is 575.99).

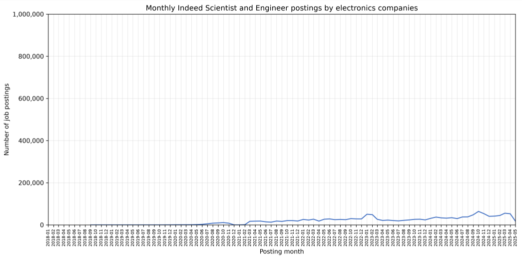
[5]This is the coarsest occupation variable, with only 7 broad categories: Engineer, Sales, Finance, Admin, Operations, Marketing, Scientist.

The number of job postings across time: Some weird patterns in the data?

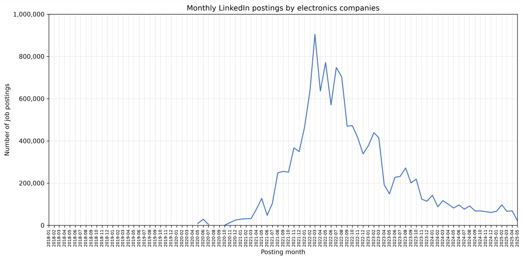
(a) Indeed, all job postings



(b) Indeed, scientists and engineers



(c) LinkedIn, all job posting



(d) LinkedIn, scientists and engineers